

DAFFODIL INTERNATIONAL UNIVERSITY

Department of Software Engineering

TEST AUTOMATION PROJECT

Test Report

Application Under Test

BigBasket Bangladesh

Testing Approach

Selenium WebDriver Based Test Automation using Page Object Model

Automation Framework

Behave (BDD)

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Group Information

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2. Project Overview

2.1 Project Name

BigBasket Bangladesh Test Automation Project

2.2 Application Under Test

BigBasket Bangladesh

2.3 Purpose of the Project

The purpose of this project is to perform functional testing of the selected web application using Selenium-based test automation and the Page Object Model approach.

The project uses Behave scenarios to describe and execute functional test cases and maintains separate page classes and step definitions as part of the automation structure.

3. Testing Objective

The main objectives of the testing project are:

- To evaluate important functional areas of the BigBasket Bangladesh web application.
- To automate repetitive functional test scenarios using Selenium WebDriver.
- To use the Page Object Model for better organization and maintainability of the automation project.
- To validate major user-facing functionalities such as homepage, login, product search, product details, categories, footer and shopping cart.
- To execute the defined test scenarios and record their results.
- To identify and document any bugs or issues observed during testing.
- To provide supporting evidence through screenshots and test execution reports.

4. Testing Scope

4.1 In Scope

The following functional areas are covered, based on the executed Behave automation report:

Module	Covered Scenarios
Homepage	Homepage loading, Homepage title, Homepage URL, Search box, Website logo
Login	Login popup, Invalid login
Product Search	Milk, Rice, Biscuit, Egg, Cooking Oil
Product Details	Product name, Product price, Product image, Product availability, Add product to cart
Shopping Cart	Add product to cart, Remove product from cart
Product Categories	Open first category, Category filters
Footer	Footer section, Footer links, About Us, Contact Us

4.2 Out of Scope

No additional out-of-scope items are assumed beyond what was implemented in the submitted automation project. Testing is limited to the functional scenarios implemented and executed in this automation project.

5. Test Execution Summary

Item	Result
Total Features	7
Passed Features	7
Failed Features	0
Total Scenarios	22
Passed Scenarios	22
Failed Scenarios	0
Total Steps	82
Passed Steps	82
Failed Steps	0
Execution Time	88.4 seconds

Overall Test Result: PASS

6. Test Result by Feature

Feature	Scenarios	Result
Shopping Cart	2	PASS
Product Categories	2	PASS
Footer Validation	4	PASS
BigBasket Bangladesh Homepage	5	PASS
Login Validation	2	PASS
Product Details	2	PASS
BigBasket Bangladesh Product Search	5	PASS
Total	22	PASS

7. Test Environment

7.1 Application Under Test

BigBasket Bangladesh Web Application

7.2 Test Type

Functional Web Automation Testing

7.3 Automation Framework

Behave — the execution report is directly Behave-generated and shows feature, scenario, and step level execution results.

7.4 Test Automation Structure

The project uses feature files with corresponding step-definition files. Example structure observed in the execution report:

- features/cart.feature
- features/homepage.feature
- features/login.feature
- features/product.feature
- features/search.feature
- features/steps/*.py (corresponding step definitions)

7.5 Test Data

The following input/search data was used across the automation scenarios:

- Milk
- Rice
- Biscuit
- Egg
- Cooking Oil
- Test credentials used for the invalid login scenario

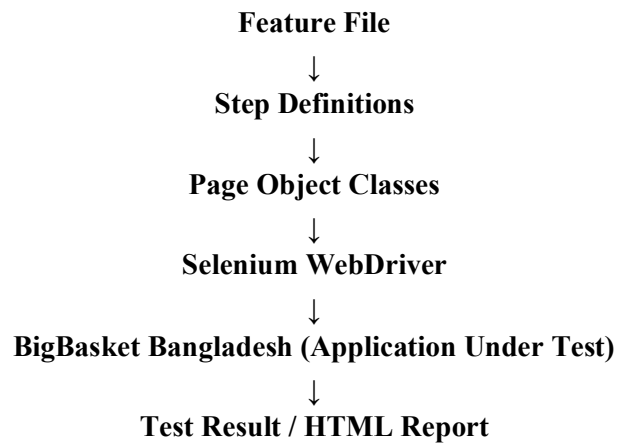
8. Tools & Technologies

Tool / Technology	Purpose
Python	Automation implementation
Selenium WebDriver	Web browser automation
Behave	BDD-based test execution
Gherkin	Feature / scenario definition
Page Object Model (POM)	Automation code organization and maintainability
HTML Test Report	Test execution results and documentation

These tools and technologies are confirmed by the project structure (feature files and Python step-definition files) and by the generated HTML execution report showing individual scenario and step mappings.

9. Automation Framework Structure

The overall automation flow followed in this project is illustrated below:



Example mapping observed in the execution report:

- features/cart.feature
- → features/steps/search_steps.py
- → features/steps/product_steps.py
- → features/steps/cart_steps.py

10. Detailed Test Cases

The table below converts all 22 executed Behave scenarios into a formal test-case format. No additional or fabricated test cases have been added.

TC ID	Module	Test Scenario	Expected Result	Actual Result	Status
TC-001	Shopping Cart	Add a product to cart	Selected product should be added to cart	Product was added to cart	PASS
TC-002	Shopping Cart	Remove a product from cart	Product should be removed and cart should be empty	Product was removed and cart became empty	PASS
TC-003	Product Categories	Open first category	Category products should be displayed	Category products displayed	PASS
TC-004	Product Categories	Verify category filters	Filter section and price filter should be displayed	Filters were displayed	PASS
TC-005	Footer	Verify footer section	Footer should be displayed	Footer displayed	PASS
TC-006	Footer	Verify footer links	Footer links should be displayed	Footer links displayed	PASS
TC-007	Footer	Open About Us page	About Us page should open successfully	Page opened successfully	PASS
TC-008	Footer	Open Contact Us page	Contact Us page should open successfully	Page opened successfully	PASS
TC-009	Homepage	Verify homepage loads successfully	Homepage should load successfully	Homepage loaded successfully	PASS
TC-010	Homepage	Verify homepage title	Homepage title should be displayed	Title displayed	PASS
TC-011	Homepage	Verify homepage URL	Homepage URL should be correct	URL verified successfully	PASS
TC-012	Homepage	Verify search box	Search box should be visible	Search box visible	PASS
TC-013	Homepage	Verify website logo	Website logo should be visible	Logo visible	PASS
TC-014	Login	Verify login popup	Email and password fields should be displayed	Required fields displayed	PASS
TC-015	Login	Invalid login	Error message should be displayed	Error message displayed	PASS
TC-016	Product Details	Verify product details	Product name, price, image and availability should be displayed	Details displayed	PASS
TC-017	Product Details	Add product to cart	Product should be added successfully	Product added successfully	PASS
TC-018	Product Search	Search for Milk	Search results should be displayed	Results displayed	PASS
TC-019	Product Search	Search for Rice	Search results should be displayed	Results displayed	PASS
TC-020	Product Search	Search for Biscuit	Search results should be displayed	Results displayed	PASS
TC-021	Product Search	Search for Egg	Search results should be displayed	Results displayed	PASS

TC ID	Module	Test Scenario	Expected Result	Actual Result	Status
TC-022	Product Search	Search for Cooking Oil	Search results should be displayed	Results displayed	PASS

Test Case Summary

Metric	Value
Total Test Cases	22
Passed	22
Failed	0
Pass Rate	100%

11. Test Evidence / Screenshots

The following screenshots (maintained in the project's screenshots folder) serve as supporting evidence for the executed test cases.

Evidence ID	Evidence Description	Related Test Case
EVD-001	Add product to cart	TC-001 / TC-017
EVD-002	Remove product from cart	TC-002
EVD-003	Invalid login	TC-015
EVD-004	Open About Us	TC-007
EVD-005	Open Contact Us	TC-008
EVD-006	Category validation	TC-003 / TC-004
EVD-007	Product search results	TC-018 – TC-022

Note: Actual screenshot file names should be inserted here exactly as saved in the project's screenshots folder before final submission.

12. Defect / Issue Summary

12.1 Automation Execution Defect Summary

Category	Count
Total Scenarios Executed	22
Passed	22
Failed	0
Automation Errors	0
Failed Steps	0

12.2 Defect Statement

Based on the recorded automation execution, no failed test scenario or failed test step was observed in the executed test suite. Therefore, no defect was raised from the recorded automation execution results.

Note: This statement does not mean the BigBasket Bangladesh application is free of defects — it only reflects the result of this specific automation execution. Any additional bugs identified through separate manual testing should be documented independently in a dedicated Bug/Issue Report.

13. Test Execution Report

The existing behave_report.html file is maintained as the supporting execution report for this test report.

Metric	Result
Features	7 passed
Scenarios	22 passed
Steps	82 passed
Execution Duration	88.4 seconds

Metric	Result
Overall Result	PASS

Supporting Report File

- reports/behave_report.html

14. Conclusion

The BigBasket Bangladesh web application was evaluated through a Selenium-based automated testing project using the defined feature scenarios and step implementations. The executed test suite covered key functional areas including the homepage, login validation, product search, product details, product categories, footer validation, and shopping cart operations.

The recorded execution completed with 7 passed features, 22 passed scenarios, and 82 passed steps, with an execution duration of 88.4 seconds. No failed scenario or failed step was recorded in this automation execution.

The test evidence and execution report are maintained as part of the project documentation. Any additional bugs or issues identified through separate manual testing should be documented independently in a detailed Bug/Issue Report.